

# What's in an inventory object

## Record (intrinsic)

### Accounting

#### Taxonomy

what life stages  
does it have

#### Life Stage

what anatomical  
forms can it take

#### Physical Form

how is it counted  
and handled

#### Size + Qty

relative to similar  
records

### Relatedness

#### Provenance type

how does it relate  
to other entries

We'll dive into this in a second...

## Modifiers (extrinsic)

### Instance

Where does this  
show up?

Which  
organization,  
site,  
(nursery)  
structure  
(tank),  
substructure  
(tray)

### Events

What's  
changed?

When was it  
created

Why did the  
quantity go up

Why did the  
health status  
change

When did it  
“graduate”

How long did it  
take to go  
from one size  
class to  
another

## Detail on Canonical “Record” Data Structure

| <b>Taxonomy</b><br>what life stages<br>does it have | <b>Life Stage</b><br>what anatomical<br>forms can it take | <b>Physical Form</b><br>how is it counted<br>and handled | <b>Size + Qty</b><br>relative to similar<br>records | <b>Provenance type</b><br>how does it relate<br>to other entries |
|-----------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------|------------------------------------------------------------------|
| coral sp                                            | gamete                                                    | container                                                | xs                                                  | founder                                                          |
| seagrass sp                                         | larvae                                                    | shared<br>substrate                                      | s                                                   | cohort                                                           |
| urchin sp                                           | recruit                                                   | individual                                               | m                                                   | sexual recruit                                                   |
| crab sp                                             | juvenile                                                  |                                                          | l                                                   |                                                                  |
| fish sp                                             | adult                                                     |                                                          | xl                                                  |                                                                  |
|                                                     | broodstock                                                |                                                          |                                                     |                                                                  |

## These are static, and don't change



| <b>Taxonomy</b><br>what life stages<br>does it have | <b>Life Stage</b><br>what anatomical<br>forms can it take | <b>Physical Form</b><br>how is it counted<br>and handled | <b>Size + Qty</b><br>relative to similar<br>records | <b>Provenance type</b><br>how does it relate<br>to other entries |
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| crab sp                                             | juvenile                                                  |                                                          | l                                                   |                                                                  |
| fish sp                                             | adult                                                     |                                                          | xl                                                  |                                                                  |
|                                                     | broodstock                                                |                                                          |                                                     |                                                                  |

# Accounting

How many/much  
have you got?

=

| These may change via events                          |                                                   |                                                      |
|------------------------------------------------------|---------------------------------------------------|------------------------------------------------------|
| Life Stage<br>what anatomical x<br>forms can it take | Physical Form<br>how is it counted<br>and handled | Size Mapping + Qty<br>relative to similar<br>records |
| gamete                                               | container                                         | xs                                                   |
| larvae                                               | shared<br>substrate                               | s                                                    |
| recruit                                              | individual                                        | m                                                    |
| juvenile                                             |                                                   | l                                                    |
| adult                                                |                                                   | xl                                                   |
| broodstock                                           |                                                   |                                                      |

# Accounting

How many/much  
have you got?

=

| determines                                   |  | determines                                             |  | determines                                        |  |                                                        |  |
|----------------------------------------------|--|--------------------------------------------------------|--|---------------------------------------------------|--|--------------------------------------------------------|--|
| Taxonomy<br>what life stages<br>does it have |  | Life Stage<br>x what anatomical x<br>forms can it take |  | Physical Form<br>how is it counted<br>and handled |  | Size Mapping + Qty<br>x relative to similar<br>records |  |
| coral sp                                     |  | gamete                                                 |  | container                                         |  | xs                                                     |  |
| seagrass sp                                  |  | larvae                                                 |  | shared<br>substrate                               |  | s                                                      |  |
| urchin sp                                    |  | recruit                                                |  | individual                                        |  | m                                                      |  |
| crab sp                                      |  | juvenile                                               |  |                                                   |  | l                                                      |  |
| fish sp                                      |  | adult                                                  |  |                                                   |  | xl                                                     |  |
|                                              |  | broodstock                                             |  |                                                   |  |                                                        |  |

# Relatedness

**Provenance type**  
(how does it relate  
to other entries)

---

**founder**

Endemic, genetically distinct individual. Collected or rescue

**cohort**

Putative genet of related individuals created from two or more parent gametes handled as a collective. Only valid for “container” and “shared substrate” physical forms

**sexual recruit**

A “graduate” of a cohort that is tracked individually for asexual propagation or experimental purposes

# Relatedness: cohort

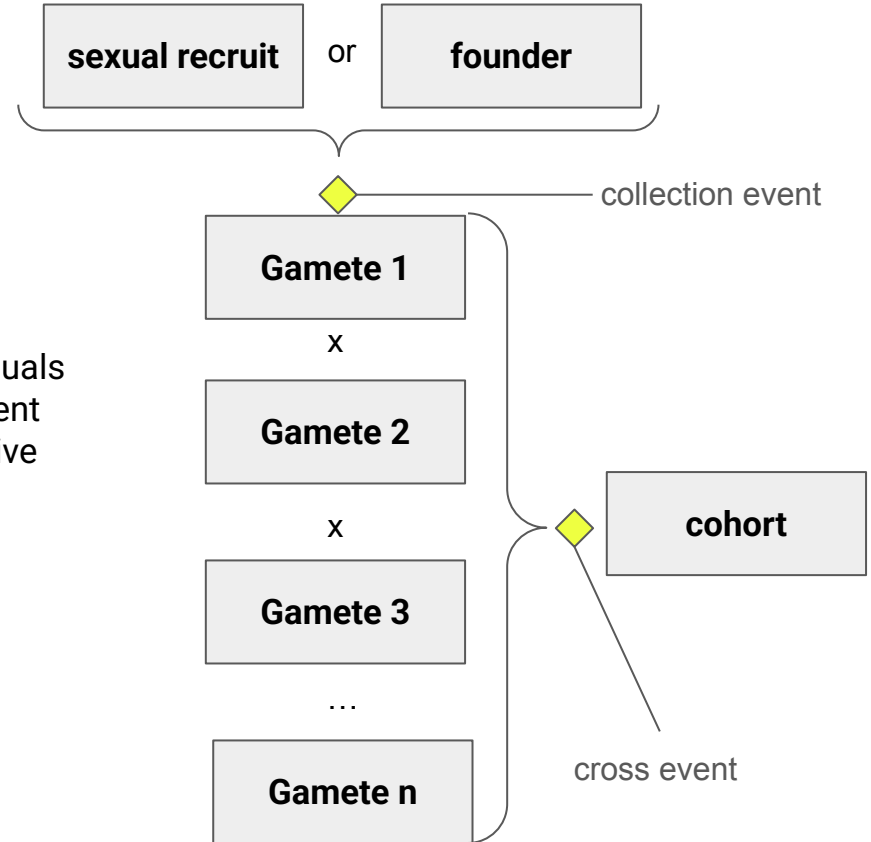
**Provenance type**  
(how does it relate  
to other entries)

**founder**

**cohort**

**sexual recruit**

Putative genet of related individuals  
created from two or more parent  
gametes handled as a collective



# Relatedness

**Provenance type**  
(how does it relate  
to other entries)

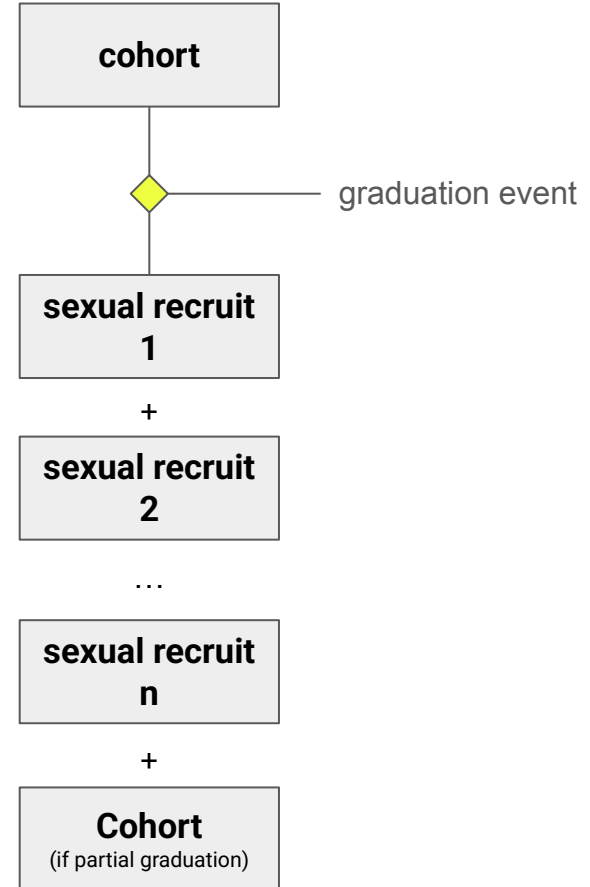
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**founder**

**cohort**

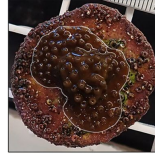
**sexual recruit**

A “graduate” of a cohort that is tracked  
individually for asexual propagation or  
experimental purposes





example:  
traditional plug



| <b>Taxonomy</b><br>what life stages<br>does it have | <b>Life Stage</b><br>what anatomical forms<br>can it take | <b>Physical Form</b><br>how is it counted and<br>handled | <b>Size + Qty</b><br>relative to similar<br>records | <b>Provenance type</b><br>how does it relate to<br>other entries |
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| coral sp                                            | gamete                                                    | container                                                | xs                                                  | founder                                                          |
| seagrass sp                                         | larvae                                                    | shared substrate                                         | s                                                   | cohort                                                           |
| urchin sp                                           | recruit                                                   | individual                                               | m                                                   | sexual recruit                                                   |
| crab sp                                             | juvenile                                                  |                                                          | l                                                   |                                                                  |
| fish sp                                             | adult                                                     |                                                          | xl                                                  |                                                                  |
|                                                     | broodstock                                                |                                                          |                                                     |                                                                  |

example:  
traditional plug



| <b>Taxonomy</b><br>what life stages<br>does it have | <b>Life Stage</b><br>what anatomical forms<br>can it take | <b>Physical Form</b><br>how is it counted and<br>handled | <b>Size + Qty</b><br>relative to similar<br>records | <b>Provenance type</b><br>how does it relate to<br>other entries |
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| fish sp                                             | adult                                                     |                                                          | xl                                                  |                                                                  |
|                                                     | broodstock                                                |                                                          |                                                     |                                                                  |

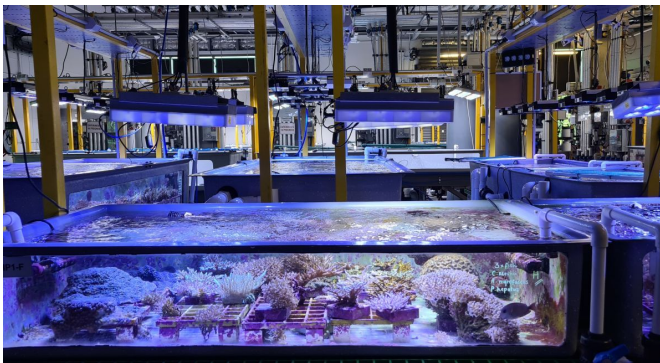
example:  
traditional clipping



note\*- we could expand “individual” to  
“mounted individual (plug)” and “unmounted  
individual” (branch) which would only be  
valid for “adult” life stage of coral

| <b>Taxonomy</b><br>what life stages<br>does it have | <b>Life Stage</b><br>what anatomical forms<br>can it take | <b>Physical Form</b><br>how is it counted and<br>handled | <b>Size + Qty</b><br>relative to similar<br>records | <b>Provenance type</b><br>how does it relate to<br>other entries |
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| crab sp                                             | juvenile                                                  |                                                          | l                                                   |                                                                  |
| fish sp                                             | adult                                                     |                                                          | xl                                                  |                                                                  |
|                                                     | broodstock                                                |                                                          |                                                     |                                                                  |

example:  
spawning colony



| <b>Taxonomy</b><br>what life stages<br>does it have | <b>Life Stage</b><br>what anatomical forms<br>can it take | <b>Physical Form</b><br>how is it counted and<br>handled | <b>Size + Qty</b><br>relative to similar<br>records | <b>Provenance type</b><br>how does it relate to<br>other entries |
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| crab sp                                             | juvenile                                                  |                                                          | l                                                   |                                                                  |
| fish sp                                             | adult                                                     |                                                          | xl                                                  |                                                                  |
|                                                     | broodstock                                                |                                                          |                                                     |                                                                  |

example:  
settled cohort on a plug

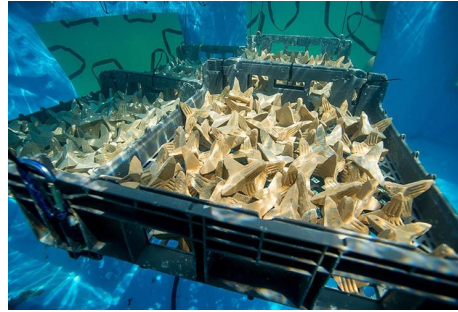


“Size” in a shared substrate is  
density (# per fixed volume)

ex:  
qty 1 xs, where xs =  
0-10 recruits per plug

| <b>Taxonomy</b><br>what life stages<br>does it have | <b>Life Stage</b><br>what anatomical forms<br>can it take | <b>Physical Form</b><br>how is it counted and<br>handled | <b>Size + Qty</b><br>relative to similar<br>records | <b>Provenance type</b><br>how does it relate to<br>other entries |
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| crab sp                                             | juvenile                                                  |                                                          | l                                                   |                                                                  |
| fish sp                                             | adult                                                     |                                                          | xl                                                  |                                                                  |
|                                                     | broodstock                                                |                                                          |                                                     |                                                                  |

example:  
settled cohort on a plug



“Size” in a shared substrate is  
density (# per fixed volume)

ex:  
qty 20 L, where L = 2k-5k  
recruits per tetrapod

| <b>Taxonomy</b><br>what life stages<br>does it have | <b>Life Stage</b><br>what anatomical forms<br>can it take | <b>Physical Form</b><br>how is it counted and<br>handled | <b>Size + Qty</b><br>relative to similar<br>records | <b>Provenance type</b><br>how does it relate to<br>other entries |
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| crab sp                                             | juvenile                                                  |                                                          | l                                                   |                                                                  |
| fish sp                                             | adult                                                     |                                                          | xl                                                  |                                                                  |
|                                                     | broodstock                                                |                                                          |                                                     |                                                                  |

example:  
settled cohort on a plug



“Size” in a shared substrate is  
density (# per fixed volume)

ex:  
qty 3 m, where m = 2-5  
juveniles per tile

| <b>Taxonomy</b><br>what life stages<br>does it have | <b>Life Stage</b><br>what anatomical forms<br>can it take | <b>Physical Form</b><br>how is it counted and<br>handled | <b>Size + Qty</b><br>relative to similar<br>records | <b>Provenance type</b><br>how does it relate to<br>other entries |
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| fish sp                                             | adult                                                     |                                                          | xl                                                  |                                                                  |
|                                                     | broodstock                                                |                                                          |                                                     |                                                                  |

example:  
Dave's Domes



“Size” in a shared substrate is  
density (# per fixed volume)

ex:  
qty 8 m, 2s where m =  
15-20 adults per dome, s  
= 10-15 adults per dome

| <b>Taxonomy</b><br>what life stages<br>does it have | <b>Life Stage</b><br>what anatomical forms<br>can it take | <b>Physical Form</b><br>how is it counted and<br>handled | <b>Size + Qty</b><br>relative to similar<br>records | <b>Provenance type</b><br>how does it relate to<br>other entries |
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| fish sp                                             | adult                                                     |                                                          | xl                                                  |                                                                  |
|                                                     | broodstock                                                |                                                          |                                                     |                                                                  |



example:  
cryopreserved sperm



“Size” in a container is  
density (# per fixed volume)

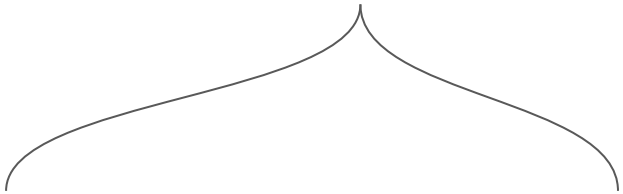
ex:  
qty 1 xs, where xs =  
500 sperm/ml

| <b>Taxonomy</b><br>what life stages<br>does it have | <b>Life Stage</b><br>what anatomical forms<br>can it take | <b>Physical Form</b><br>how is it counted and<br>handled | <b>Size + Qty</b><br>relative to similar<br>records | <b>Provenance type</b><br>how does it relate to<br>other entries |
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| crab sp                                             | juvenile                                                  |                                                          | l                                                   |                                                                  |
| fish sp                                             | adult                                                     |                                                          | xl                                                  |                                                                  |
|                                                     | broodstock                                                |                                                          |                                                     |                                                                  |

example:  
Not possible

X

“Container” cannot be applied to “broodstock”. “Cohort” can only be associated with container and shared substrate Physical Forms.



| <b>Taxonomy</b><br>what life stages<br>does it have | <b>Life Stage</b><br>what anatomical forms<br>can it take | <b>Physical Form</b><br>how is it counted and<br>handled | <b>Size + Qty</b><br>relative to similar<br>records | <b>Provenance type</b><br>how does it relate to<br>other entries |
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| crab sp                                             | juvenile                                                  |                                                          | l                                                   |                                                                  |
| fish sp                                             | adult                                                     |                                                          | xl                                                  |                                                                  |
|                                                     | broodstock                                                |                                                          |                                                     |                                                                  |